

Program 9

Step 1:- Start minikube

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: storage-provisioner, dashboard, default-storageclass
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2:- Create namespaces.yaml

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ cat namespaces.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

Step 3:- Create pods.yaml

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ cat pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

Step 4:- Apply namespaces and pods

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
```

Step 5:- Verify namespaces

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl get namespaces
NAME                STATUS   AGE
default             Active   12h
dev                 Active   26s
kube-node-lease     Active   12h
kube-public         Active   12h
kube-system         Active   12h
kubernetes-dashboard Active   12h
prod                Active   26s
```

Step 6:- Verify pods by namespaces

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl get pods -n dev
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           43s
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl get pods -n prod
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           49s
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl get pods
No resources found in default namespace.
```

Step 7:- Filter pods by namespaces

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS    RESTARTS   AGE
frontend-pod  1/1     Running   0           79s
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS    RESTARTS   AGE
backend-pod   1/1     Running   0           97s
```

Step 8:- Cleanup- Remove all pods and namespaces

```
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl delete pods all -n dev
Error from server (NotFound): pods "all" not found
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl delete pods --all -n dev
pod "frontend-pod" deleted from dev namespace
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl delete pods --all -n prod
pod "backend-pod" deleted from prod namespace
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl delete namespace dev
namespace "dev" deleted
1rv24mc024_chandan_athreya_s@JARVIS:~/Devops-LabSessions/prog9$ kubectl delete namespace prod
namespace "prod" deleted
```